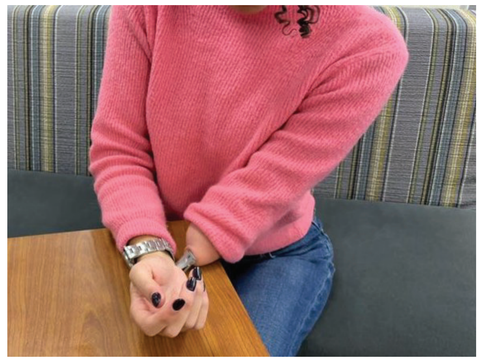
# Hands-Free Watchband

Isaac Weaver, Carter Stentzel, Brian Jensen, Nathan Usevitch
Brigham Young University, Provo, Utah, 84602, USA

## Background



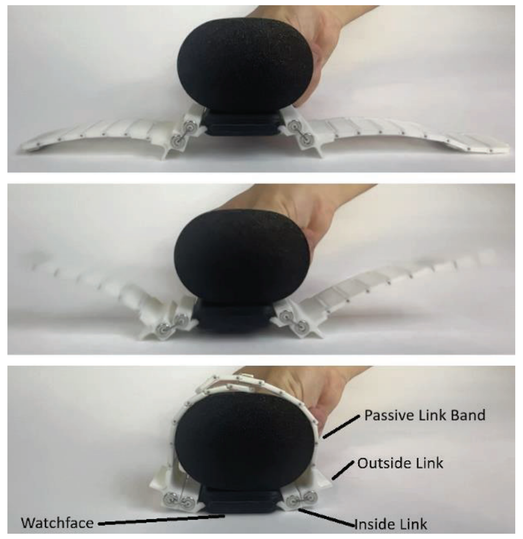
Conventional watchband

Objective: Create a watchband that a user can put on and take off using only one hand to improve accessibility.

Magnet band

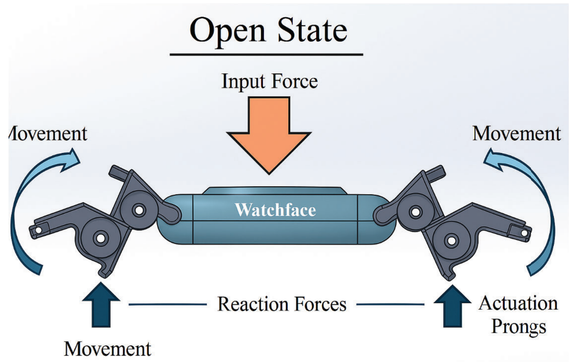
Scissor band

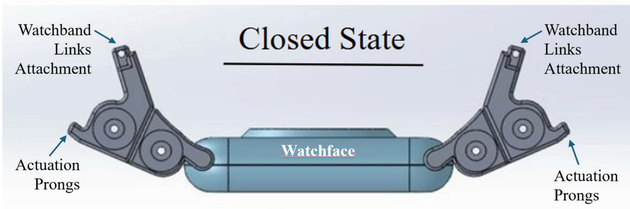
## Magnet Band Design

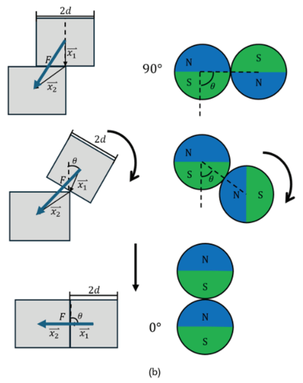


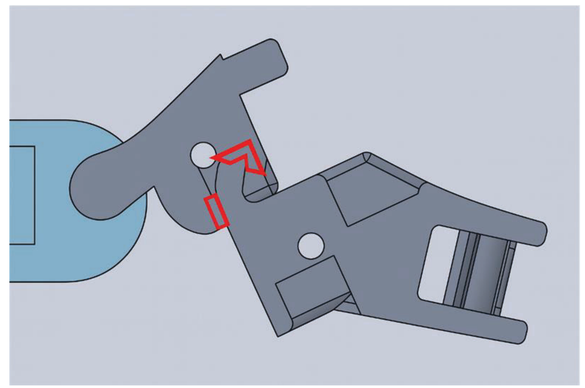

Still shots of watchband actuating on wrist dummy

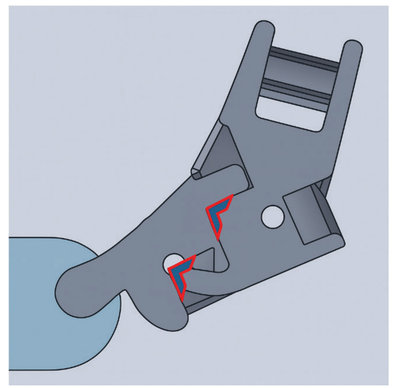
The band relies on a new mechanism employing magnets. The magnetic fields interact to create two stable positions at open and closed with smooth transitions between the two when pushed down



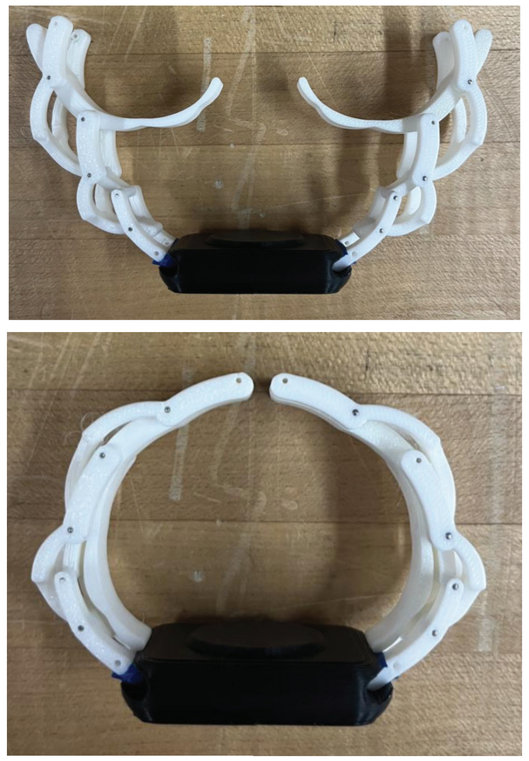



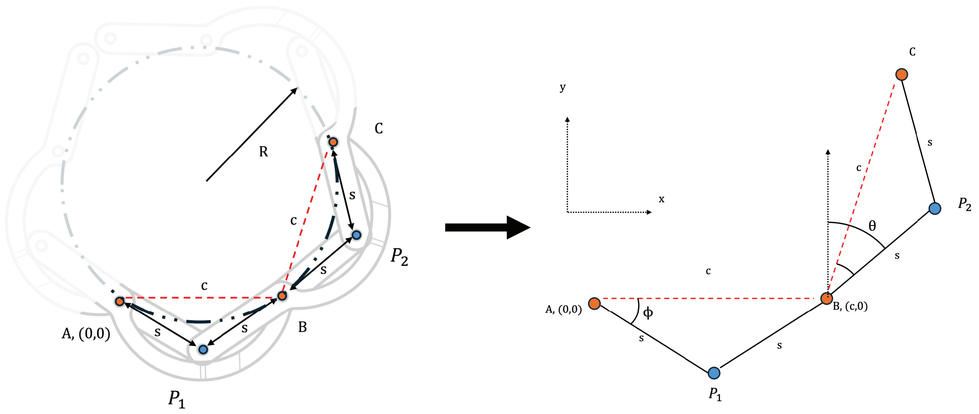




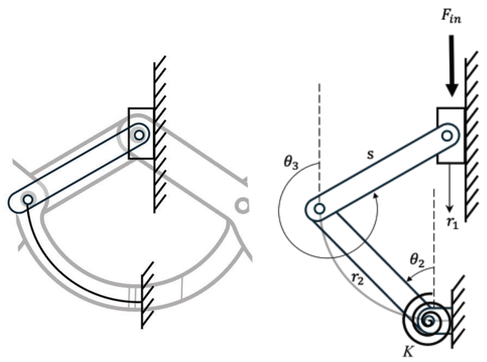
Section view of mechanism showing hard stops used to create the take-off track highlighted in red.

## Compliant Scissor Band



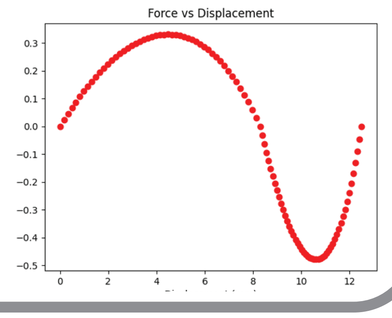


The scissor band uses a novel compliant scissor link to create a bi stable band. The band is actuated by pushing the wrist down on the lever arms. The band stores energy until the unstable equilibrium.Then it snaps through into the second stable position releasing the stored energy and closing.



$$F_{in} = \frac{K(\theta_2 - \theta_{2_o})cos(\theta_3)}{r_2 sin(\theta_2 - \theta_3)}$$

Each link is modeled as an initially curved compliant beam with a pin and slider. This results in the force displacement curve of a bi-stable mechanism.




Acknowledgments

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